

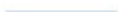
SQL and Analytical Data Modelling

Transform analytical data from relational storage to distributed pipelines.



Learning outcomes

01  Write joins, aggregations, windows, and CTEs.

02  Choose grain and keys before metrics.

03  Test queries for duplication and missingness.

The concept system

01

joins

02

window functions

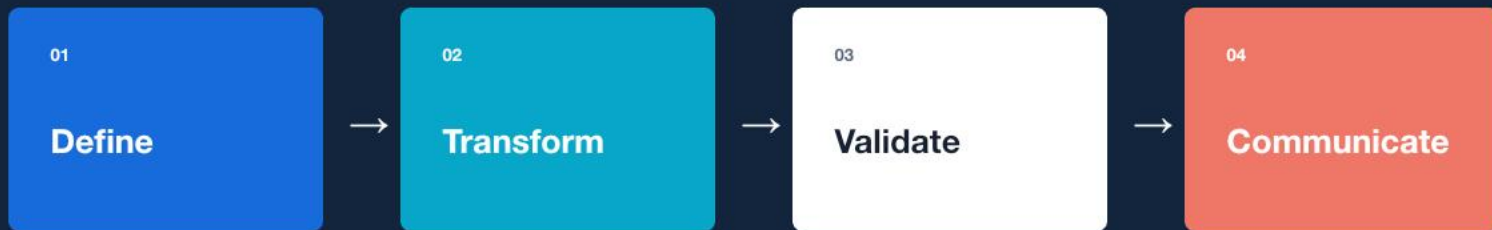
03

grain

04

dimensional modelling

A repeatable reasoning loop



Model orders and customers, then compute retention and revenue.

Guided lab

LAB BRIEF

**Model orders and customers,
then compute retention and
revenue.**

workbook / tests / evidence

01 **Frame**

Define inputs, outputs, and success criteria.

02 **Build**

Implement the smallest correct workflow.

03 **Verify**

Test normal, boundary, and failure cases.

04 **Explain**

Record the evidence and remaining limits.

Portfolio deliverable

A documented SQL model with data tests.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Analytical Pipeline at Two Scales

Data model 25%, SQL 25%,
Spark 25%, tests 15%,
documentation 10%.

What should remain after the lesson?

CONCEPT

joins + window functions

PRACTICE

Model orders and customers, then compute retention and revenue.

EVIDENCE

A documented SQL model with data tests.
